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Temperature control structure and temperature control method for transport container

Abstract

A temperature control structure for a transport container includes: a temperature control compartment provided in a truck; a transport container that is loaded into the temperature control compartment, and has a box shape capable of accommodating an object, in which at least a part of a wall constituting the box shape is a heat conduction portion made of metal; and a heat exchanger of a solid-state heat transfer type that is disposed in contact with the heat conduction portion and controls a temperature of the object via the heat conduction portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a national phase application in the United States of International Patent Application No. PCT/JP2022/018316 with an international filing date of Apr. 20, 2022, which claims priority of Japanese Patent Application No. 2021-092340 filed on Jun. 1, 2021 and Japanese Patent Application No. 2022-035577 filed on Mar. 8, 2022, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a temperature control structure and a temperature control method for a transport container.

BACKGROUND ART

(3) A temperature control transport vehicle that transports an object while controlling temperature is known. Generally, in the temperature control transport vehicle, an object is accommodated in a transport container made of resin, and the transport container is loaded and transported in a temperature control compartment provided in a luggage compartment of such as a truck. The inside of the temperature control compartment is temperature-controlled by an air conditioner, and the object is temperature-controlled.

(4) For example, JP 2004-12059 A discloses a freezing and refrigerating chamber temperature control vehicle as an example of a temperature control transport vehicle. In the freezing and refrigerating chamber temperature control vehicle of JP 2004-12059 A, temperature management is performed in two chambers of freezing and refrigeration by using only one cold storage plate.

SUMMARY OF THE INVENTION

(5) In the freezing and refrigerating chamber temperature control vehicle of JP 2004-12059 A, a cooling structure of a convection heat transfer type that controls an air temperature in a cooling compartment to a low temperature is adopted. Therefore, it is necessary to manage the entire room in the cooling compartment at a low temperature regardless of the size and number of the transport containers. Therefore, there is room for improvement in cooling efficiency.

(6) An object of the present invention is to improve temperature control efficiency as compared with temperature control of a convection heat transfer type in a temperature control structure and a temperature control method for a transport container.

(7) A first aspect of the present invention provides a temperature control structure for a transport container, the temperature control structure including: a temperature control compartment provided in a transport machine; a transport container that is loaded into the temperature control compartment, and has a box shape capable of accommodating an object, in which at least a part of a wall constituting the box shape is a heat conduction portion made of metal; and a heat exchanger of a solid-state heat transfer type that is disposed in contact with the heat conduction portion and controls a temperature of the object via the heat conduction portion.

(8) According to this configuration, the temperature control structure of a solid-state heat transfer type that directly controls the temperature of the transport container by the heat exchanger is adopted. Therefore, in the heat exchanger, the refrigerant at the required temperature and the transport container exchange heat via the heat conduction portion, and the temperature of the transport container can be controlled. In the temperature control structure of a solid-state heat transfer type, since necessary temperature control can be realized according to the size and the number of the transport containers, the temperature control efficiency can be improved as compared with the temperature control of the convection heat transfer type. In addition, in the

transport container, the heat conduction portion made of metal is provided at a contact portion with the heat exchanger. Therefore, it is possible to exhibit high heat exchange efficiency as compared with a material having a low thermal conductivity such as a resin. Note that, in the transport container, the entirety including the other portion of the heat conduction portion may be made of metal.

(9) When the plurality of transport containers are controlled to different temperatures by the convection heat transfer method, it is necessary to partition the temperature control compartment with a partition wall because the temperature of the air needs to be changed for each required temperature. However, in the case of controlling the plurality of transport containers to different temperatures in a solid-state heat transfer manner, it is also possible to control the plurality of transport containers to respective required temperatures by arranging a plurality of heat exchangers using refrigerants having different temperatures in one room without partitioning the inside of the temperature control compartment by the partition wall. Since convection heat is also transferred to the surrounding air on the surface of the transport container, the temperature control compartment may be partitioned into a plurality of rooms by the partition wall as a means for further increasing the temperature control efficiency, and a plurality of heat exchangers using refrigerants having different temperatures may be arranged for the plurality of rooms, respectively.

(10) According to these configurations, the temperature of the object can be controlled according to various required temperatures. Accordingly, the objects having different required temperatures can be transported at a time. For example, the temperature control compartment may be divided into a plurality of rooms corresponding to required temperatures of freezing, refrigeration, normal temperature, and high temperature.

(11) The transport container may have a plurality of protruding portions having a heat insulating property on a first surface that is an outer surface.

(12) According to this configuration, when the transport container comes into contact with an arbitrary surface having no temperature control function, it is possible to suppress the heat entering/exiting from the surface to the transport container. For example, even when the transport container is placed on an arbitrary floor surface having no temperature control function, the temperature of the transport container (that is, the temperature of the object) can be maintained at the required temperature.

(13) The transport container may have a plurality of recessed portions having shapes complementary to the protruding portions on a second surface that is an outer surface facing the first surface.

(14) According to this configuration, the transport containers can be arranged by combining the recessed portions and the protruding portions, and the states of the transport containers can be stabilized. For example, the transport containers can be stacked so as to stabilize the states.

(15) A plurality of the transport containers may be loaded in contact with each other in the temperature control compartment, and at least a part of a portion where the transport containers are in contact with each other may be the heat conduction portion.

(16) According to this configuration, since the thermal energy from the heat exchanger is also transferred between the plurality of transport containers, efficient temperature control can be realized.

(17) The heat conduction portion may be made of an aluminum alloy.

(18) According to this configuration, since the aluminum alloy has a higher thermal conductivity and a lower radiation factor than other common metals, efficient temperature control can be realized.

(19) The temperature control structure for a transport container may further include a cold storage material or a heat storage material that controls a temperature of the transport container.

(20) According to this configuration, the temperature of the transport container can be further controlled by the cold storage material or the heat storage material.

(21) A second aspect of the present invention provides a temperature control method for a transport container, the temperature control method including: preparing a temperature control compartment provided in a transport machine, a transport container that is loaded into the temperature control compartment, and has a box shape capable of accommodating an object, in which at least a part of a wall constituting the box shape is a heat conduction portion made of metal, and a heat exchanger that controls a temperature of the object; and bringing the heat exchanger and the heat conduction portion into contact with each other, and controlling the temperature of the object by the heat exchanger in a solid-state heat transfer manner via the heat conduction portion.

(22) According to this method, the temperature control method of a solid-state heat transfer type that directly controls the temperature of the transport container by the heat exchanger is adopted. Therefore, in the heat exchanger, the refrigerant at the required temperature and the transport container exchange heat via the heat conduction portion, and the temperature of the transport container can be controlled. In the temperature control method of a solid-state heat transfer type, since necessary temperature control can be realized according to the size and the number of the transport containers, the temperature control efficiency can be improved as compared with the temperature control of the convection heat transfer type. In addition, in the transport container, the heat conduction portion made of metal is provided at a contact portion with the heat exchanger. Therefore, it is possible to exhibit high heat exchange efficiency as compared with a material having a low thermal conductivity such as a resin. Note that, in the transport container, the entirety including the other portion of the heat conduction portion may be made of metal.

(23) According to the present invention, since the temperature control structure of a solid-state heat transfer type is adopted, it is possible to provide the temperature control structure and the temperature control method for a transport container with improved temperature control efficiency as compared with the temperature control of the convection heat transfer type.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic configuration diagram of a truck in which a temperature control structure for a transport container according to a first embodiment of the present invention is adopted;
- (2) FIG. 2 is a perspective view of the transport container;
- (3) FIG. 3 is a lower perspective view of a transport container in a first modification;
- (4) FIG. 4 is an upper perspective view of a transport container in a second modification;
- (5) FIG. 5 is a lower perspective view of the transport container in the second modification;
- (6) FIG. 6 is an upper perspective view of a transport container in a third modification;
- (7) FIG. 7 is a lower perspective view of a transport container in a fourth modification;
- (8) FIG. 8 is a lower perspective view of a transport container in a fifth modification;
- (9) FIG. 9 is an upper perspective view of a transport container in a sixth modification;
- (10) FIG. 10 is a broken perspective view of a transport container in a seventh modification;
- (11) FIG. 11 is a schematic configuration diagram of a truck in which a temperature control structure for a transport container in another modification is adopted; and
- (12) FIG. 12 is a schematic configuration diagram of a truck in which a temperature control structure for a transport container according to a second embodiment is adopted.

DETAILED DESCRIPTION

(13) Hereinafter, as embodiments of the present invention, a temperature control structure for a transport container for transporting an object while controlling the temperature will be described with reference to the accompanying drawings. As the temperature control structure, a cooling structure is exemplified in the first embodiment, and a heating structure is exemplified in the second embodiment. In the following description, terms indicating directions, such as up or down,

may be used, which shall refer to directions in a state where the transport container is loaded onto a transport machine.

First Embodiment

(14) FIG. 1 shows a schematic configuration diagram of a truck **1** in which a temperature control structure for a transport container **10** according to the first embodiment is adopted. For example, the truck **1** transports an object (cooling object in the present embodiment) such as a frozen food, a lunch box, a rice ball, or a sandwich at a low temperature from a low-temperature warehouse such as a frozen center or a chilled center located in each place to a destination such as a convenience store all over the country.

(15) The truck **1** is an example of a transport machine, and the temperature control structure of the present disclosure can be applied to any transport machine such as other vehicles or ships. In addition, the object is not particularly limited, and any object requiring temperature control other than those described above is targeted, and the transport source and the transport destination may be various.

(16) Referring to FIG. 1, a temperature control compartment **20** is disposed in the truck **1**. The temperature control compartment **20** is located behind a driver's cab **3** of the truck **1**. The temperature control compartment **20** includes a box-shaped outer wall **21** and a partition wall **22** disposed inside the box-shaped outer wall **21**. The inside of the temperature control compartment **20** is partitioned into a plurality of rooms (two rooms R1 and R2 in the present embodiment) by the partition wall **22**. Preferably, the outer wall **21** and the partition wall **22** have thermal insulation performance in order to increase temperature control efficiency.

(17) In the present embodiment, for each required temperature of the object, one room R1 is provided for freezing, and the other room R2 is provided for refrigeration. For example, the required temperature of the object may be about -20°C . in a room R1 for freezing, and about 5°C . in a room R2 for refrigerating. Although not illustrated in detail, a room for room temperature may be further provided by adding a partition wall **22**. For example, in the room for normal temperature, the required temperature of the object may be about 20°C . When it is not necessary to transport the object for each temperature or when the temperature difference between a plurality of objects having different required temperatures is not large, the entire inside of the temperature control compartment **20** may be provided as one room without providing the partition wall **22**.

(18) A plurality of transport containers **10** are loaded into the temperature control compartment **20**. In the temperature control compartment **20**, the plurality of transport containers **10** are loaded in contact with each other in the horizontal direction and the vertical direction. Details of the transport container **10** will be described later.

(19) The temperature control structure of the present disclosure is of a solid-state heat transfer type using heat exchangers **30** and **40**. That is, the heat exchangers **30** and **40** are in contact with at least a part of the transport container **10**, and the heat exchangers **30** and **40** directly control the temperature of the transport container **10** (cool the transport container **10** in the present embodiment).

(20) In the present embodiment, the heat exchangers **30** and **40** are of a plate type, for example, and are provided for the plurality of rooms R1 and R2, respectively. The heat exchangers **30** and **40** are disposed to be exposed to an inner lower surface of the temperature control compartment **20**. The heat exchangers **30** and **40** are fluidly connected to compressors **32** and **42** and capacitors **33** and **43**, respectively, via pipes **31** and **41** (schematically indicated by broken lines). Refrigerant flows in the pipes **31** and **41**. The refrigerant is compressed by the compressors **32** and **42**, liquefied by the capacitors **33** and **43**, and evaporated by the heat exchangers **30** and **40**. That is, the heat exchangers **30** and **40** function as evaporators. In this manner, the refrigerant circulates between the heat exchangers **30** and **40**, the compressors **32** and **42**, and the capacitors **33** and **43** via the pipes **31** and **41**, respectively. That is, a general refrigeration cycle is configured.

(21) The heat exchangers **30** and **40** have temperature control surfaces **30a** and **40a** having

temperature control functions, respectively. The temperature control surfaces **30a** and **40a** are disposed so as to be exposed to the inner lower surface of the temperature control compartment **20**. Therefore, the user can control the temperatures of the transport containers **10** (cool the transport containers **10** in the present embodiment) by placing the transport containers **10** on the temperature control surfaces **30a** and **40a**. Alternatively, the temperature control surfaces **30a** and **40a** of the heat exchangers **30** and **40** may be disposed on an inner side surface, an inner upper surface, or the like of the temperature control compartment **20**.

(22) FIG. 2 shows a perspective view of the transport container **10**. In FIG. 2, the front of the truck **1** is indicated by an X direction, the side is indicated by a Y direction, and the upper side is indicated by a Z direction. The same applies to the following drawings.

(23) The transport container **10** has a box shape capable of accommodating an object **2** (schematically indicated by a broken line). For example, the outer shape of the transport container **10** is a rectangular parallelepiped shape. In the present embodiment, the transport container **10** is made of an aluminum alloy. Therefore, the entire outer wall of the transport container **10** is a heat conduction portion **11** having a higher thermal conductivity than the resin. The heat conduction portion **11** of the transport container **10** is disposed on the temperature control surfaces **30a** and **40a** (see FIG. 1) of the heat exchangers **30** and **40**, and is directly temperature-controlled from the temperature control surfaces **30a** and **40a** of the heat exchangers **30** and **40**. In particular, silicon grease or the like having a high thermal conductivity may be applied to the heat conduction portion **11**. In addition, in the present embodiment, since the entire outer wall of the transport container **10** is the heat conduction portion **11**, it is possible to dispose the transport container **10** on the temperature control surfaces **30a** and **40a** with an arbitrary surface of the transport container **10** facing downward. Alternatively, not the entire transport container **10** but a part thereof may be used as the heat conduction portion **11**, and the heat conduction portion **11** may be disposed on the temperature control surfaces **30a** and **40a** (see FIG. 1) of the heat exchangers **30** and **40**. In addition, the material of the heat conduction portion **11** is not limited to the aluminum alloy, and may be other metals.

(24) In addition, the opening and closing structure of the transport container **10** for accommodating the object **2** is not particularly limited, and may be any structure. For example, as illustrated in FIG. 2, the transport container **10** may have a main body **12** that accommodates the object **2** and a lid body **13** that closes the main body **12**. In addition, the lid body **13** may be pivotally supported by a hinge **14** with respect to the main body **12**. In addition, the hinge **14** and the lid body **13** can be configured at arbitrary positions and shapes. For example, as illustrated in FIG. 2, a structure in which the upper side of the transport container **10** is opened may be used, or alternatively, a structure in which the side is opened and closed may be used. Further, a structure in which the lid body **13** is detachable from the main body **12** without the hinge **14** may be used.

(25) According to the present embodiment, the following operational effects are obtained.

(26) A temperature control structure of a solid-state heat transfer type is adopted in which the temperature of the transport container **10** is directly controlled (cooled in the present embodiment) by the heat exchangers **30** and **40**. Therefore, in the heat exchangers **30** and **40**, the low-temperature refrigerant and the transport container **10** exchange heat via the heat conduction portion, and the temperature of the transport container **10** can be controlled. In the temperature control structure of a solid-state heat transfer type, since necessary temperature control can be realized according to the size and the number of the transport containers **10**, the temperature control efficiency can be improved as compared with the temperature control of the convection heat transfer type. In addition, in the transport container **10**, the heat conduction portion **11** made of metal is provided at a contact portion with the heat exchangers **30** and **40**. Specifically, in the present embodiment, the entire transport container **10** is the heat conduction portion **11** made of metal. Therefore, it is possible to exhibit high heat exchange efficiency as compared with a material having a low thermal conductivity such as a resin.

(27) As described above, the inside of the temperature control compartment **20** is partitioned into the two rooms **R1** and **R2** by the partition wall **22**. The heat exchangers **30** and **40** using refrigerants having different temperatures are disposed in the two rooms **R1** and **R2**, respectively. Therefore, the temperature of the object **2** can be controlled according to the required temperature. Accordingly, the objects **2** having different required temperatures can be transported at a time.

(28) In addition, the plurality of transport containers **10** are loaded in contact with each other in the up-down direction and the left-right direction in the temperature control compartment **20**. In the present embodiment, since the entire outer wall of the transport container **10** is the heat conduction portion **11** made of an aluminum alloy, the cold energy obtained by the transport container **10** arranged at the lowermost stage from the heat exchangers **30** and **40** is efficiently transmitted to the other transport containers **10**. Therefore, it is possible to realize efficient temperature control of the solid-state heat transfer type among the plurality of transport containers **10**. Alternatively, at least a part of a portion where the transport containers **10** are in contact with each other may be used as the heat conduction portion **11** instead of the entire outer wall of the transport container **10**.

(29) The transport container **10** is made of an aluminum alloy. Since the aluminum alloy has a higher thermal conductivity and a lower radiation factor than other common metals, efficient temperature control can be realized.

(30) (First Modification)

(31) FIG. **3** shows a lower perspective view of a transport container **10** in a first modification. In FIG. **3**, illustration of the opening/closing structure of the transport container **10** is omitted. In addition, in FIG. **4** and subsequent drawings, illustration of the opening/closing structure of the transport container **10** is similarly omitted.

(32) The transport container **10** in the present modification has a plurality of protruding portions **15** having a heat insulating property on a lower surface **10a** (illustrated as an upper surface in FIG. **3**). Here, the lower surface **10a** is an example of a first surface of the present invention. The number of the plurality of protruding portions **15** is six in the present modification, and one of them is denoted by a reference sign. The plurality of protruding portions **15** are made of resin. The resin is preferable as the material of the plurality of protruding portions **15** because it has a lower thermal conductivity than metal and a heat insulating property. More preferably, the plurality of protruding portions **15** are made of foamed resin having higher heat insulating properties. The plurality of protruding portions **15** are bonded to the lower surface **10a** of the transport container **10** with an adhesive. In the example of FIG. **3**, each of the plurality of protruding portions **15** has a rectangular parallelepiped shape, but the shape is not particularly limited. Further, the plurality of protruding portions **15** may be provided on the side surface of the transport container **10**.

(33) According to the present modification, when the transport container **10** is placed on an arbitrary floor surface having no temperature control function, heat transfer between the floor surface and the transport container **10** can be suppressed. In other words, even when the transport container **10** is placed on an arbitrary floor surface having no temperature control function, the temperature of the transport container **10** (that is, the temperature of the object **2**) can be maintained at the required temperature.

(34) (Second Modification)

(35) FIG. **4** shows a lower perspective view of a transport container **10** in a second modification. FIG. **5** shows an upper perspective view of the transport container **10** in the second modification.

(36) As in the first modification, the transport container **10** in the present modification has a plurality of protruding portions **15** having a heat insulating property on a lower surface **10a** (illustrated as an upper surface in FIG. **5**). The number of the plurality of protruding portions **15** is nine in the present modification, and one of them is denoted by a reference sign (see FIG. **5**). The plurality of protruding portions **15** are formed by molding the lower surface **10a** into a protruding shape by press molding. Preferably, the plurality of protruding portions **15** are formed by bonding, for example, resin members **15a** (see hatched portions) having a low thermal conductivity to the

molded surface having a protruding shape. In the present modification, each of the plurality of protruding portions **15** has a truncated cone shape.

(37) In addition, the transport container **10** in the present modification has a plurality of recessed portions **16** having shapes complementary to the plurality of protruding portions **15** on an upper surface **10b**. Here, the upper surface **10b** is an example of a second surface of the present invention. The number of the plurality of recessed portions **16** is nine in the present modification, and one of them is denoted by a reference sign (see FIG. **4**). That is, each of the plurality of recessed portions **16** in the present modification is a recess having a truncated cone shape. In addition, the plurality of recessed portions **16** may be provided on a side surface facing a side surface on which the plurality of protruding portions **15** are provided in the transport container **10**.

(38) When the transport containers **10** are arranged in the up-down direction, the plurality of protruding portions **15** are arranged so as to be accommodated in the plurality of recessed portions **16**. At this time, the upper surface **10b** of one transport container **10** and the lower surface **10a** of the other transport container **10** come into contact with each other, and thermal energy is exchanged between heat exchangers **30** and **40**.

(39) According to the present modification, the states of the transport containers **10** can be stabilized by stacking the transport containers **10** by combining the plurality of recessed portions **16** and the plurality of protruding portions **15**.

(40) (Third Modification)

(41) FIG. **6** shows an upper perspective view of a transport container **10** in a third modification.

(42) Similarly to the first and second modifications, the transport container **10** in the present modification has a plurality of (three in the present modification) protruding portions **15** having a heat insulating property on a lower surface **10a**. The plurality of protruding portions **15** are formed by molding the lower surface **10a** into an elongated protruding shape (protrusion) by press molding. Preferably, the plurality of protruding portions **15** are formed by bonding resin members **15a** (see hatched portions) having a low thermal conductivity to the molded protrusion surface.

(43) In addition, the transport container **10** in the present modification has a plurality of (three in the present modification) recessed portions **16** having shapes complementary to the plurality of protruding portions **15** on an upper surface **10b**. Each of the plurality of recessed portions **16** is formed in a groove shape on the upper surface **10b**.

(44) When the transport containers **10** are arranged in the up-down direction, the plurality of protruding portions **15** are arranged so as to be accommodated in the plurality of recessed portions **16**. At this time, the plurality of protruding portions **15** are slidable in the plurality of recessed portions **16**. Therefore, the operation of stacking the transport containers **10** can be facilitated.

(45) (Fourth Modification)

(46) FIG. **7** shows a lower perspective view of a transport container **10** in a fourth modification.

(47) The transport container **10** in the present modification has a shape in which the plurality of protruding portions **15** in the third modification are tapered downward (illustrated as upward in FIG. **7**).

(48) According to the present modification, the ground contact area of the plurality of protruding portions **15** can be reduced. Therefore, it is possible to reduce heat conduction with the floor surface when the transport container **10** is placed on an arbitrary floor surface having no temperature control performance.

(49) FIG. **8** shows a lower perspective view of a transport container **10** in a fifth modification.

(50) Similarly to the fourth modification illustrated in FIG. **7**, the transport container **10** in the present modification has a shape in which a plurality of protruding portions **15** are tapered downward (illustrated as upward in FIG. **8**).

(51) In addition, in the present modification, a plurality of recessed portions **16** of the transport container **10** have shapes complementary to the plurality of protruding portions **15**. That is, the plurality of recessed portions **16** also have a shape tapered downward (illustrated as upward in FIG.

8).

(52) According to the present modification, the contact area between the plurality of protruding portions **15** and the plurality of recessed portions **16** can be increased, and the heat conduction efficiency can be enhanced.

(53) (Sixth Modification)

(54) FIG. **9** shows an upper perspective view of a transport container **10** in a sixth modification.

(55) On a lower surface **10a** of the transport container **10** in the present modification, a plurality of protruding portions **15** same as those in the third modification are formed. An upper surface **10b** is provided with recessed portions **16** in a lattice pattern.

(56) When the transport containers **10** are arranged in the up-down direction, the plurality of protruding portions **15** are arranged so as to be accommodated in the recessed portions **16** in a lattice pattern. At this time, the plurality of protruding portions **15** are slidable in the recessed portions **16** in a lattice pattern. Therefore, the operation of stacking the transport containers **10** can be facilitated.

(57) FIG. **10** shows a broken perspective view of a transport container **10** in a seventh modification.

(58) On a lower surface **10a** of the transport container **10** in the present modification, a plurality of protruding portions **15** same as those in the third modification are formed. A cold storage material **17** for temperature control is disposed inside each of the plurality of protruding portions **15**. As the cold storage material **17**, one having various temperature control performances can be used according to the required temperature. For example, a resin member **15a** (see a hatched portion) having a low thermal conductivity is bonded to each of the surfaces of the plurality of protruding portions **15**.

(59) According to the present modification, the temperature of the transport container **10** can be further controlled by the cold storage materials **17**. In particular, since the cold storage material **17** is disposed in each of the plurality of protruding portions **15** and the resin member **15a** (see the shaded portion) is bonded to each of the surfaces of the plurality of protruding portions **15**, it is possible to reduce the amount of heat input from the floor surface when the transport container **10** is placed on an arbitrary floor surface having no temperature control performance. The arrangement of the cold storage material **17** is not particularly limited in the plurality of protruding portions **15**, and may be any position where the temperature of the transport container **10** can be controlled.

(60) (Other Modification)

(61) FIG. **11** shows a schematic configuration diagram of a truck **1** in which a temperature control structure for a transport container **10** in another modification is adopted.

(62) In the present modification, the partition wall **22** (see FIG. **1**) in the above embodiment is omitted. That is, the inside of a temperature control compartment **20** includes one room **R3**. In the one room **R3**, heat exchangers **30** and **40** using refrigerants having different temperatures are disposed.

(63) According to the present modification, unlike the convection heat transfer type in which the temperature control compartment **20** needs to be partitioned by the partition wall in the case of transporting objects having different required temperatures, the temperature control structure of a solid-state heat transfer type is adopted, so that objects having different required temperatures can be transported at once without partitioning the temperature control compartment **20** into a plurality of rooms.

Second Embodiment

(64) FIG. **11** shows a schematic configuration diagram of a truck **1** in which a temperature control structure for a transport container **10** according to the second embodiment is adopted. For example, the truck **1** transports an object (heating object in the present embodiment) **2** at a high temperature. The description of the same configuration as that of the first embodiment may be omitted.

(65) Also in the present embodiment, the truck **1** is an example of a transport machine, and the

temperature control structure of the present disclosure can be applied to any transport machine such as other vehicles or ships. In addition, the object **2** is not particularly limited, and any object requiring temperature control is targeted, and the transport source and the transport destination may be various.

(66) In the present embodiment, for each required temperature of an object **2**, one room **R1** is provided for heating at a high temperature, and the other room **R2** is provided for heating at a higher temperature than the one room. Although not illustrated in detail, a room for room temperature may be further provided by adding a partition wall **22**. When it is not necessary to transport the object **2** for each temperature or when the temperature difference between a plurality of objects having different required temperatures is not large, the entire inside of a temperature control compartment **20** may be provided as one room without providing the partition wall **22**.

(67) A plurality of transport containers **10** are loaded into the temperature control compartment **20**. In the temperature control compartment **20**, the plurality of transport containers **10** are loaded in contact with each other in the horizontal direction and the vertical direction. The transport container **10** is the same as that of the first embodiment.

(68) The temperature control structure of the present disclosure is of a solid-state heat transfer type using heat exchangers **30** and **40**. That is, the heat exchangers **30** and **40** are in contact with at least a part of the transport container **10**, and the heat exchangers **30** and **40** directly control the temperature of the transport container **10** (heat the transport container **10** in the present embodiment).

(69) In the present embodiment, the heat exchangers **30** and **40** are of a plate type, for example, and are provided for the plurality of rooms **R1** and **R2**, respectively. The heat exchangers **30** and **40** are disposed to be exposed to an inner lower surface of the temperature control compartment **20**. The heat exchangers **30** and **40** are fluidly connected to compressors **32** and **42** and evaporators **34** and **44**, respectively, via pipes **31** and **41** (schematically indicated by broken lines). Refrigerant flows in the pipes **31** and **41**. The refrigerant is compressed by the compressors **32** and **42**, is condensed by the heat exchangers **30** and **40**, heats the object **2**, and evaporates by the evaporators **34** and **44**. That is, in the present embodiment, the heat exchangers **30** and **40** function as capacitors. In this manner, the refrigerant circulates between the compressors **32** and **42**, the heat exchangers **30** and **40**, and the evaporators **34** and **44** via the pipes **31** and **41**, respectively.

(70) The heat exchangers **30** and **40** have temperature control surfaces **30a** and **40a** having temperature control functions, respectively. The temperature control surfaces **30a** and **40a** are disposed so as to be exposed to the inner lower surface of the temperature control compartment **20**. Therefore, the user can control the temperatures of the transport containers **10** (heat the transport containers **10** in the present embodiment) by placing the transport containers **10** on the temperature control surfaces **30a** and **40a**. Alternatively, the temperature control surfaces **30a** and **40a** of the heat exchangers **30** and **40** may be disposed on an inner side surface, an inner upper surface, or the like of the temperature control compartment **20**.

(71) The structure of the transport container **10** illustrated in FIGS. **2** to **10** is also applicable in the present embodiment. As a result, thermal energy is efficiently transferred among the plurality of transport containers **10**. The cold storage material **17** for cooling in FIG. **10** is replaced with a heat storage material **17** for heating.

(72) Although specific embodiments of the present invention and modifications thereof have been described above, the present invention is not limited to the above embodiments, and various modifications can be made within the scope of the present invention. For example, an appropriate combination of the contents of individual modifications may be an embodiment of the present invention. That is, a temperature control structure having both a cooling structure and a heating structure may be provided.

(73) In the above embodiment, substantially the entire transport container **10** is made of an aluminum alloy, but at least a part of a contact portion with the heat exchangers **30** and **40** and a

portion where the transport containers **10** are in contact with each other may be made of metal (for example, made of an aluminum alloy) as the heat conduction portion **11**. In addition, the transport container **10** may have a heat insulating material, a cold storage material, or a heat storage material on the outer surface other than the heat conduction portion **11** from the viewpoint of maintaining the required temperature. In addition to the temperature control structure of a solid-state heat transfer type of the above embodiment, an air conditioner that controls the temperature in the temperature control compartment **20** in a convection heat transfer manner may be provided. That is, the temperature control structures of both the convection heat transfer type and the solid-state heat transfer type may be adopted. **1** truck **2** object **3** driver's cab **10** transport container **10a** lower surface (first surface) **10b** upper surface (second surface) **11** heat conduction portion **12** main body **13** lid body **14** hinge **15** protruding portion **15a** resin member **16** recessed portion **17** cold storage material (heat storage material) **20** temperature control compartment **21** outer wall **22** partition wall **30** heat exchanger **30a** temperature control surface **31** pipe **32** compressor **33** capacitor **34** evaporator **40** heat exchanger **40a** temperature control surface **41** refrigerant pipe **42** compressor **43** capacitor **44** evaporator **R1, R2, R3** room

Claims

1. A temperature control structure for a transport container, the temperature control structure comprising: a temperature control compartment provided in a transport machine; a transport container that is loaded into the temperature control compartment, and has a box shape capable of accommodating an object, in which at least a part of a wall constituting the box shape is a heat conduction portion made of metal; and a heat exchanger of a solid-state heat transfer type that is disposed in contact with the heat conduction portion and controls a temperature of the object via the heat conduction portion.
2. The temperature control structure for a transport container according to claim 1, wherein a plurality of the transport containers are loaded in contact with each other in the temperature control compartment, and at least a part of a portion where the transport containers are in contact with each other is the heat conduction portion.
3. The temperature control structure for a transport container according to claim 1, wherein the heat conduction portion is made of an aluminum alloy.
4. The temperature control structure for a transport container according to claim 1, further comprising a cold storage material or a heat storage material that controls a temperature of the transport container.
5. The temperature control structure for a transport container according to claim 1, wherein the transport container has a plurality of protruding portions having a heat insulating property on a first surface that is an outer surface.
6. The temperature control structure for a transport container according to claim 5, wherein the transport container has a plurality of recessed portions having shapes complementary to the protruding portions on a second surface that is an outer surface facing the first surface.
7. The temperature control structure for a transport container according to claim 1, wherein an inside of the temperature control compartment is partitioned into a plurality of rooms, and the heat exchangers using refrigerants having different temperatures are arranged for the plurality of rooms, respectively.
8. The temperature control structure for a transport container according to claim 7, wherein the transport container has a plurality of protruding portions having a heat insulating property on a first surface that is an outer surface.
9. The temperature control structure for a transport container according to claim 8, wherein the transport container has a plurality of recessed portions having shapes complementary to the protruding portions on a second surface that is an outer surface facing the first surface.

10. The temperature control structure for a transport container according to claim 7, wherein a plurality of the transport containers are loaded in contact with each other in the temperature control compartment, and at least a part of a portion where the transport containers are in contact with each other is the heat conduction portion.
 11. The temperature control structure for a transport container according to claim 7, wherein the heat conduction portion is made of an aluminum alloy.
 12. The temperature control structure for a transport container according to claim 7, further comprising a cold storage material or a heat storage material that controls a temperature of the transport container.
 13. The temperature control structure for a transport container according to claim 1, wherein the temperature control compartment includes one room, and a plurality of the heat exchangers using refrigerants having different temperatures are disposed in the one room.
 14. The temperature control structure for a transport container according to claim 13, wherein the transport container has a plurality of protruding portions having a heat insulating property on a first surface that is an outer surface.
 15. The temperature control structure for a transport container according to claim 14, wherein the transport container has a plurality of recessed portions having shapes complementary to the protruding portions on a second surface that is an outer surface facing the first surface.
 16. The temperature control structure for a transport container according to claim 13, wherein a plurality of the transport containers are loaded in contact with each other in the temperature control compartment, and at least a part of a portion where the transport containers are in contact with each other is the heat conduction portion.
 17. The temperature control structure for a transport container according to claim 13, wherein the heat conduction portion is made of an aluminum alloy.
 18. The temperature control structure for a transport container according to claim 13, further comprising a cold storage material or a heat storage material that controls a temperature of the transport container.
 19. A temperature control method for a transport container, the temperature control method comprising: preparing a temperature control compartment provided in a transport machine, a transport container that is loaded into the temperature control compartment, and has a box shape capable of accommodating an object, in which at least a part of a wall constituting the box shape is a heat conduction portion made of metal, and a heat exchanger that controls a temperature of the object; and bringing the heat exchanger and the heat conduction portion into contact with each other, and controlling the temperature of the object by the heat exchanger in a solid-state heat transfer manner via the heat conduction portion.
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